PROJECT PROPOSAL

Visa Tracking and Application System

VTAS

Proposal Phase 4

Author: Group 34

Name	Surname	Email
Eunice	Tshisuaka	402413721@my.richfield.ac.za
Jacques	Fredericks	402313271@my.richfield.ac.za
Teague	Pillay	402311013@my.richfield.ac.za
Pierre-Albert	van der Merwe	402311257@my.richfield.ac.za
Jeremiah	Senkgane	402411132@my.richfield.ac.za
Daniel	Hurrell	402313208@my.richfield.ac.za

4.1 Introduction 4
4.2 Systems Design (Description of the Proposed System) 5
4.2.1 Proposed System Overview 5
4.2.2 Architectural Design 5
4.2.3 Main System Modules 6
4.2.4 Processing and Data Flow 6
4.2.5 Physical and Deployment Design 6
4.3 Architectural Design 8
4.3.1 Software Architectural Design 8
4.3.2 Hardware Architectural Design 9
4.3.3 Network Architectural Design 9
4.3.4 Class Diagram 10
4.4 Physical Design 11
4.4.2 Hardware Requirements 11
4.4.3 Software Requirements 12
4.4.4 Network and Communication Requirements 12
4.4.5 Storage and File Management 13
4.4.6 Physical design overview 14
4.5. Database Design VTAS 15
4.5.1. Main database tables 15
4.5.2. Table structure 15
4.5.3. Relationships between tables 19
4.5.4. Basic database rules 19
4.6 Program Design (Program Pseudo Code) 21
Introduction for Section 4.6 21
4.6.1 User Registration 21
4.6.2 User Login 22
4.6.3 Submit Visa Application 22
4.6.4 Upload Supporting Documents 23
4.6.5 Track Visa Application Status 24

4.6.6 Visa Officer Review Process 25
4.6.7 Notification Process 26
4.6.8 Generate Reports 26
4.7 Interface Design 28
4.7.1 Menu Interface Design 28
4.7.2 Input Interface Design 28
4.7.3 Output Interface Design 29
4.8 Security and Backup Design (Software Concerns) 30
4.8.1 Application Security Design 30
4.8.2 Backup Design 31
4.8.3 Recovery and Continuity 31
4.8.4 Key Software Security and Backup Concerns 32
References 33

4.1 Introduction

The system design phase explains the anticipated Visa Tracking and Application System (VTAS) and how it will work. In previous phase round the team has experienced challenges in the visa application procedure in South Africa such as delays, manual record keeping, difficulty in tracking the progress of applications and the like hood that visitors would not know the expiry date of their visa. The observed flaws show that there is a need for a more systematic and trustworthy way of managing information on visas.

VTAS is designed to be a centralized system that allows applicants to submit visa applications and follow the status of their application, while allowing administrators to manage applicant information and update the progress of applications. The system would also assist in keeping track of visa expiry dates and sending notifications before the visa expires. It aims to cut down on manual effort, minim-e errors, enhance communication, and enable visitors to take appropriate actions before their visa expires.

The suggested system design will be described in this phrase. The design will cover the software, hardware and network architecture, physical design, database structure, program logic, user interfaces, and security and backup mechanisms. The designs will provide a clear roadmap for how the system should operate and how information on applicants would be kept, accessed and protected. The goal is to establish a safe, reliable and user-friendly system for a more structured visa application and tracking system in South Africa. (POPIA) (Republic of South Africa, 2013).

4.2 Systems Design (Description of the Proposed System)

The proposed Visa Tracking and Application System (VTAS) is a web-based system designed to provide applicants and authorised administrators with one central platform for managing visa applications, passport details, visa records, application statuses and expiry reminders. The design replaces fragmented manual records with a structured process in which information is captured once, validated, stored securely and made available to users according to their assigned access rights. (OWASP, 2025).

4.2.1 Proposed System Overview

VTAS will support two main user groups. Applicants will register, sign in, maintain their profiles, capture passport and visa information, submit or track applications, view the number of days remaining before expiry and receive reminders. Administrators will review applicant records, update application statuses, monitor approaching expiry dates, manage authorised records and review system activity. The system will not approve or extend visas independently; official decisions will remain the responsibility of the relevant immigration authority. Role-based permissions align system access with assigned organisational roles (NIST, n.d.).

4.2.2 Architectural Design

- **Presentation layer:** Responsive web pages will provide registration, login, applicant dashboards, administrative dashboards and forms for capturing and viewing records.
- **Application layer:** Node.js services will process business rules, validate information, calculate remaining visa validity, control status changes and schedule notifications.
- **Data layer:** A relational database will store users, applicants, passports, visa applications, visas, status history, notifications and audit logs.
- **Integration layer:** Controlled application programming interfaces may exchange approved information with immigration, border-control, biometric and messaging services.
- **Security layer:** Authentication, role-based access control, encryption, input validation, secure sessions and audit logging will apply across all layers.

4.2.3 Main System Modules

Module	Description
User and Access Management	Registers users, authenticates login attempts, manages profiles and applies applicant or administrator permissions.
Passport and Visa Management	Captures, validates, updates and displays passport, visa and supporting application information.
Application Tracking	Records the current application stage and maintains a history of status changes for transparency and follow-up.
Expiry Monitoring	Calculates remaining validity and identifies active, nearing-expiry and expired passports or visas.
Notification Management	Schedules and records reminders, including proposed 90-day, 30-day and 7-day expiry notifications.
Dashboard and Reporting	Provides users with relevant summaries, alerts and application information according to their roles.
Audit and Administration	Logs important activity and enables authorised administrators to monitor records and system use.

4.2.4 Processing and Data Flow

A user will access VTAS through a browser and authenticate before using protected functions. Submitted information will be checked for completeness, valid formats, duplicate identifiers and logical dates before it is processed. Approved transactions will be written to the database and relevant events will be recorded in the audit log. When an application status or expiry condition changes, the application layer will update the dashboard and create the required notification. Responses will then be returned to the user interface without exposing database credentials or internal processing details.

4.2.5 Physical and Deployment Design

The proposed deployment will separate the web interface, application services and database to reduce direct exposure of sensitive records. Users will connect through an encrypted HTTPS connection. The application server will execute the Node.js logic and communicate with the database through a restricted service account. The database will not be directly accessible from the public network. Development, testing and production environments should be separated so that test activity and sample data cannot affect live information.

Separating public-facing services from protected data stores supports defence-in-depth and controlled access (OWASP, 2025).

4.3 Architectural Design

4.3.1 Software Architectural Design

- **Layered Architecture**
  - Presentation Layer: Web portal and mobile app interfaces for applicants and administrators.
  - Application Layer: Business logic handling visa expiry tracking, notifications, and application status updates.
  - Data Layer: Database management system (MySQL) storing applicant, visa, and notification records.
  - Security Layer: Authentication, authorization, and encryption mechanisms ensuring compliance with POPIA.
  - *Extended Analysis:* Layered architecture improves modularity, scalability, and maintainability. Each layer can evolve independently, reducing coupling.
- **Client–Server Model**
  - Clients: Applicants accessing via browser/mobile.
  - Server: Centralized application server hosting VTAS logic.
  - *Extended Analysis:* This model ensures centralized control, easier updates, and consistent data integrity across all users.
- **Notification Subsystem**
  - Automated alerts via SMS/email at 90, 30, and 7 days before expiry.
  - Dashboard alerts for administrators.
  - *Extended Analysis:* Improves transparency and reduces accidental overstays.
- **Security Features**
  - Role-based access control (visitor vs. admin).
  - Data encryption for sensitive fields (passport numbers, visa IDs).
  - *Extended Analysis:* Protects against unauthorized access and ensures compliance with South African data laws.

4.3.2 Hardware Architectural Design

- **Server Infrastructure**
  - Centralized server hosting VTAS application.
  - Backup servers for redundancy and disaster recovery.
  - *Extended Analysis:* Ensures high availability and reliability of visa tracking services.
- **Client Devices**
  - Applicants: Smartphones, tablets, laptops.
  - Administrators: Desktop PCs with secure access.
  - *Extended Analysis:* Cross-platform compatibility ensures inclusivity for tech-savvy and non-tech-savvy users.
- **Database Hardware**
  - Dedicated storage servers for MySQL databases.
  - RAID configurations for fault tolerance.
  - *Extended Analysis:* Supports scalability as visa applications increase with tourism growth.
- **Networking Hardware**
  - Routers, firewalls, and load balancers.
  - *Extended Analysis:* Ensures secure, efficient traffic flow and prevents bottlenecks.

4.3.3 Network Architectural Design

- **Topology**
  - Star topology connecting clients to a central server.
  - *Extended Analysis:* Simplifies management and isolates failures.
- **Protocols**
  - HTTPS for secure communication.
  - SMTP for email notifications.
  - *Extended Analysis:* Ensures encrypted communication and reliable delivery of alerts.
- **Firewall & VPN**
  - Protects against cyber threats.

    - VPN access for administrators working remotely.
    - *Extended Analysis:* Enhances security and supports flexible work arrangements.
- **Scalability**
    - Cloud integration for handling peak loads.
    - *Extended Analysis:* Supports future expansion as tourism increases.

4.3.4 Class Diagram

- **Core Classes**
    - User: Attributes include userID, name, email, role.
    - Visa: Attributes include visaID, type, issueDate, expiryDate.
    - Application: Attributes include applicationID, status, submissionDate.
    - Notification: Attributes include notificationID, type, dateSent.
    - *Extended Analysis:* Each class represents a real-world entity, ensuring logical mapping between system and visa processes.
- **Relationships**
    - User ↔ Visa (One-to-Many): A user can hold multiple visas.
    - Visa ↔ Application (One-to-One): Each visa is linked to a specific application.
    - Application ↔ Notification (One-to-Many): Multiple notifications can be generated per application.
    - *Extended Analysis:* Relationships ensure data integrity and traceability.
- **Inheritance**
    - AdminUser inherits from User with additional privileges.
    - VisitorUser inherits from User with limited access.
    - *Extended Analysis:* Supports role-based access control.

4.4 Physical Design

VTAS will provide a single platform where users can submit visa applications, upload supporting documents, track application progress and receive notifications regarding application status changes and visa expiry dates. The system will use JavaScript for application development and MySQL as its relational database management system. The application will be hosted locally, which allows users to access the system within the configured local environment.

4.4.2 Hardware Requirements

Client devices: Users will access the application through desktop computers, laptops or other compatible devices with a web browser. The only requirements are an internet connection to be able to access the platform.

Application Server: A server will act as the host for the web application and its associated services. It will run the application, process requests from users and facilitate communication with the MySQL database. The host will have sufficient resources to host all the services

Storage capacity: Adequate disk space will be required for the application files, MySQL database, uploaded supporting documents and operational logs. Storage requirements will increase as additional applications and documents are added to the system.

4.4.3 Software Requirements

VTAS will use a combination of client-side technologies, server-side software and database software to provide its functionality.

JavaScript: JavaScript will be used to implement application functionality. On the client side, it will support user interactions and perform form validation.

HTML and CSS: HTML will define the structure of the web pages, while CSS will control the layout, styling and presentation of the user interface. These technologies will work with JavaScript to provide an interactive web application.

Node.js: Node.js will provide the runtime environment for executing server-side JavaScript. It will allow the application to process requests, apply validation rules and interact with MySQL through an appropriate database driver or library (Node.js, 2024).

MySQL: MySQL will be used as the relational database management system. It will store structured information required by our application, including user accounts, visa application records, application statuses, document metadata and notification records. Relationships and constraints will help maintain data integrity.

Web browser: A modern web browser, such as Google Chrome, Microsoft Edge or Mozilla Firefox, will be used to access the application.

4.4.4 Network and Communication Requirements

VTAS will initially operate within a local hosting environment. The application will be accessed through a web browser using the address and port configured on the host computer.

Where the application is accessed from the same computer on which it is hosted, the loopback address may be used. Where access from other devices on the same local network is required, the host computer's local IP address and the appropriate application port will be used.

The application server will communicate with the MySQL database to retrieve, insert and update information. Database access will be restricted to the application and authorized administrative tools. MySQL should not be exposed directly to client devices.

The local network must provide reliable connectivity for users who access the application from other devices. Firewall rules will be configured to permit only the connections required for the system to function.

4.4.5 Storage and File Management

VTAS will manage structured database records and supporting documents submitted as part of visa applications.

Database storage: MySQL will store structured information such as user details, application information, status history and notification records. This information will be persisted in the database so that users can retrieve their application details when they return to the system.

Document storage: Supporting documents uploaded by applicants will be stored in a designated directory on the local host computer. MySQL will store the relevant metadata and file references required to associate each document with its corresponding visa application.

File management: The system will validate uploaded files by checking permitted file types and maximum file sizes. Documents will be assigned appropriate storage references, and access will be controlled through the application rather than allowing users to browse the storage directory directly.

Storage monitoring: The available disk space will be monitored to ensure that uploaded documents and database files do not exceed the capacity of the local host. As the system grows, additional storage may be required.

The local storage environment will be suitable for development and demonstration. However, the use of a single local computer creates a potential point of failure, making appropriate backup arrangements important.

4.4.6 Physical design overview

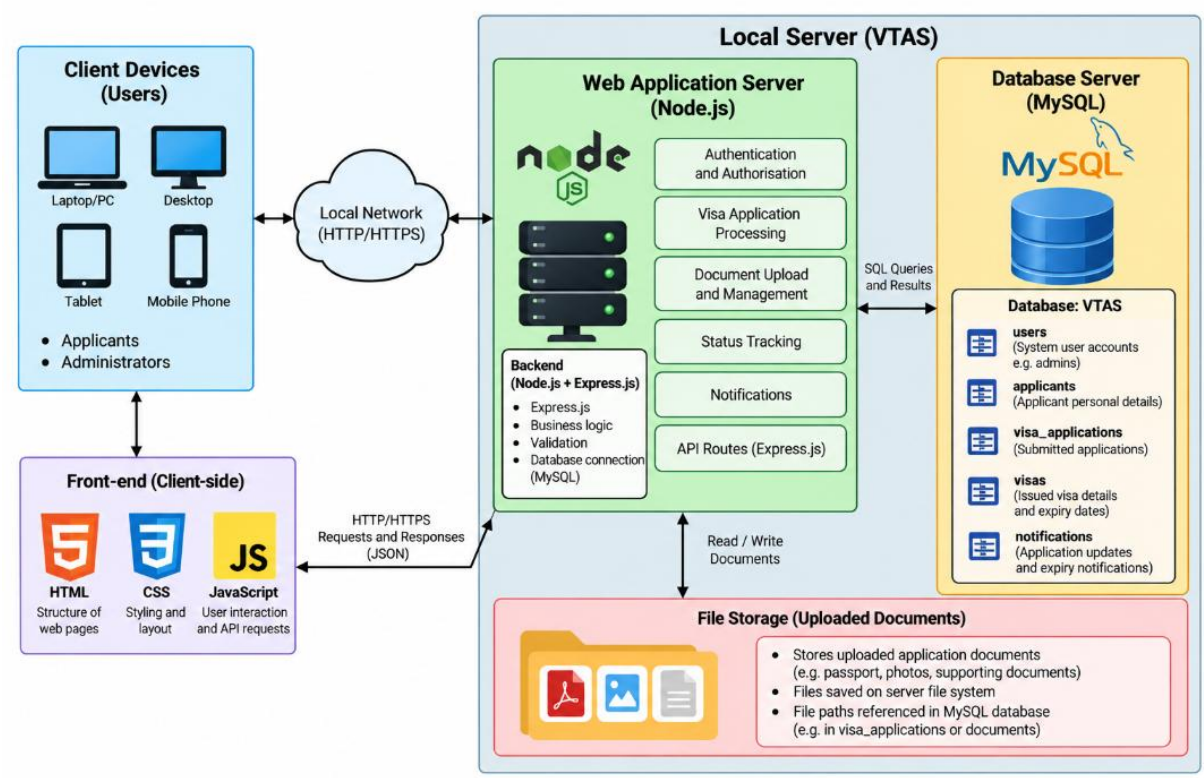


Figure 1: Overview of software systems used.

4.5. Database Design VTAS

4.5.1. Main database tables

Table	Purpose
Users	Stores user account and login details.
Applicants	Stores the applicant's personal information and passport details.
Visa_Applications	Stores visa applications and their current status.
Visas	Stores information about approved visas and their expiry dates.
Notifications	Stores reminders and updates sent to users.

4.5.2. Table structure

Here are the basic fields each table would contain.

Users

Field	Purpose
user_id (PK)	Unique user ID
email	User's email address

password	Hashed password
role	Applicant or administrator

Applicants

Field	Purpose
applicant_id (PK)	Unique applicant ID
user_id (FK)	Links to the user account
full_name	Applicant's full name
date_of_birth	Applicant's date of birth
nationality	Applicant's nationality
passport_number	Applicant's passport number
passport_expiry	Passport expiry date

Visa_Applications

Field	Purpose
application_id (PK)	Unique application ID

applicant_id (FK)	Links to the applicant
visa_type	Type of visa requested
destination_country	Country being visited
application_date	Date of application
status	Current application status

Visas

Field	Purpose
visa_id (PK)	Unique visa ID
application_id (FK)	Links to the application
visa_number	Issued visa number
issue_date	Visa issue date
expiry_date	Visa expiry date

Notifications

Field	Purpose

notification_id (PK)	Unique notification ID
user_id (FK)	Recipient of the notification
message	Notification message
notification_date	Date the notification is scheduled or created
is_sent	Indicates whether the notification has been sent

4.5.3. Relationships between tables

Users

user_id (PK)

Applicants

applicant_id (PK), user_id (FK)

Visa_Applications

application_id (PK), applicant_id (FK)

Visas

visa_id (PK), application_id (FK)

Notifications

notification_id (PK), user_id (FK)

Simplified conceptual diagram. Notifications are linked to users independently.

In simple terms:

- A user has an applicant profile.
- An applicant can submit multiple visa applications.
- An approved application can have a visa record.
- A user can receive multiple notifications.

4.5.4. Basic database rules

To keep the database reliable, we only need a few important rules:

- Each table must have a primary key.
- Each user must have a unique email address.
- Required fields cannot be left empty.

- Visa expiry dates must be later than issue dates.
- Applications must use predefined statuses, such as Submitted, Under Review, Approved or Rejected.
- Applicants must only be able to view their own information.

4.6 Program Design (Program Pseudo Code)

Introduction for Section 4.6

Program design defines the logical processes that will be implemented in the Visa Tracking and Application System. The pseudocode shows the sequence of operations for the core system functions identified during the analysis phase. These functions include user registration, user authentication, visa application submission, document management, application tracking, application review, notifications, and report generation. The pseudocode provides a clear representation of the system logic before the development stage begins.

4.6.1 User Registration

BEGIN User Registration

INPUT Name

INPUT Surname

INPUT Email

INPUT Phone Number

INPUT Password

IF Email does not exist THEN

SAVE user details

DISPLAY "Registration Successful"

ELSE

DISPLAY "Email Already Exists"

ENDIF

END User Registration

4.6.2 User Login

BEGIN User Login

INPUT Email

INPUT Password

SEARCH User Database

IF Email and Password are valid THEN

GRANT Access

OPEN User Dashboard

ELSE

DISPLAY "Invalid Login Credentials"

ENDIF

END User Login

4.6.3 Submit Visa Application

BEGIN Visa Application

INPUT Applicant Information

SELECT Visa Type

INPUT Travel Details

IF all mandatory fields are completed THEN

GENERATE Application ID

SET Status = "Pending"

SAVE Application

DISPLAY "Application Submitted Successfully"

ELSE

DISPLAY "Please Complete Required Fields"

ENDIF

END Visa Application

4.6.4 Upload Supporting Documents

BEGIN Upload Documents

SELECT Passport Copy

SELECT Photo

SELECT Additional Documents

FOR each uploaded file

IF file type is valid THEN

SAVE file

ELSE

DISPLAY "Invalid File Type"

ENDIF

NEXT

DISPLAY "Documents Uploaded Successfully"

END Upload Documents

4.6.5 Track Visa Application Status

BEGIN Track Application

INPUT Application ID

SEARCH Application Database

IF Application Exists THEN

DISPLAY Current Status

DISPLAY Application Progress

ELSE

DISPLAY "Application Not Found"

ENDIF

END Track Application

4.6.6 Visa Officer Review Process

BEGIN Review Application

SELECT Pending Application

VERIFY Applicant Details

VERIFY Supporting Documents

IF Documents are Complete THEN

SET Status = "Approved"

ELSE

SET Status = "Rejected"

ENDIF

SAVE Status

SEND Notification To Applicant

END Review Application

4.6.7 Notification Process

BEGIN Notification

WHEN Application Status Changes

RETRIEVE Applicant Email

GENERATE Notification Message

SEND Notification

LOG Notification Activity

END Notification

4.6.8 Generate Reports

BEGIN Generate Report

SELECT Report Type

RETRIEVE Application Records

PROCESS Data

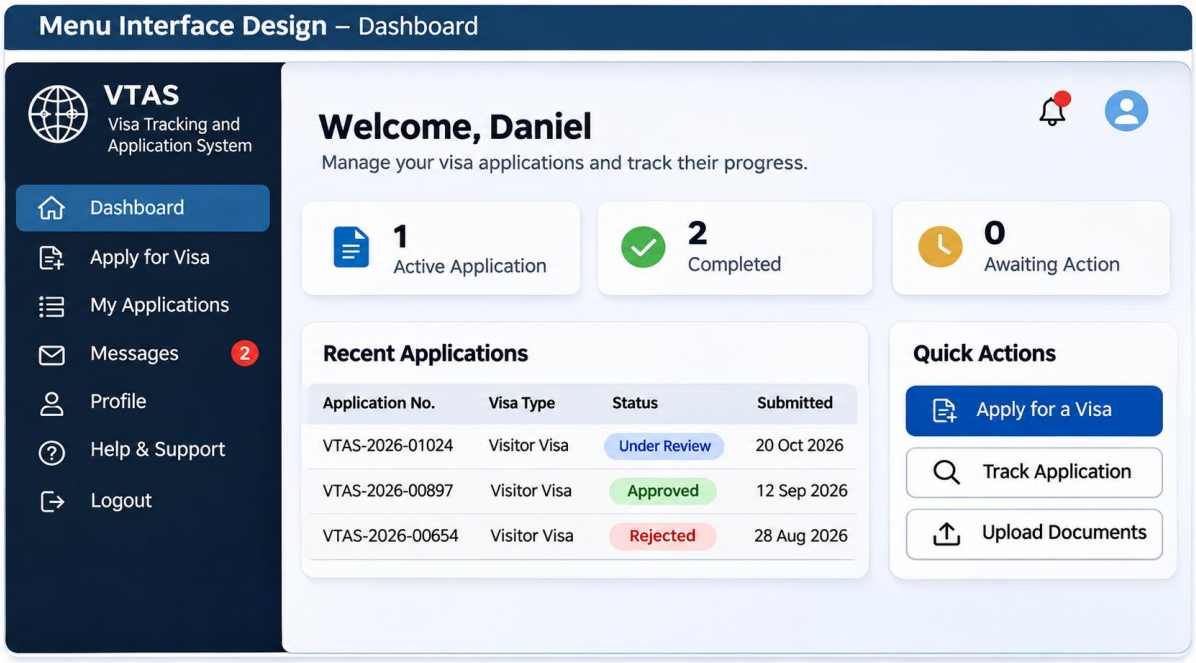
GENERATE Report

DISPLAY Report

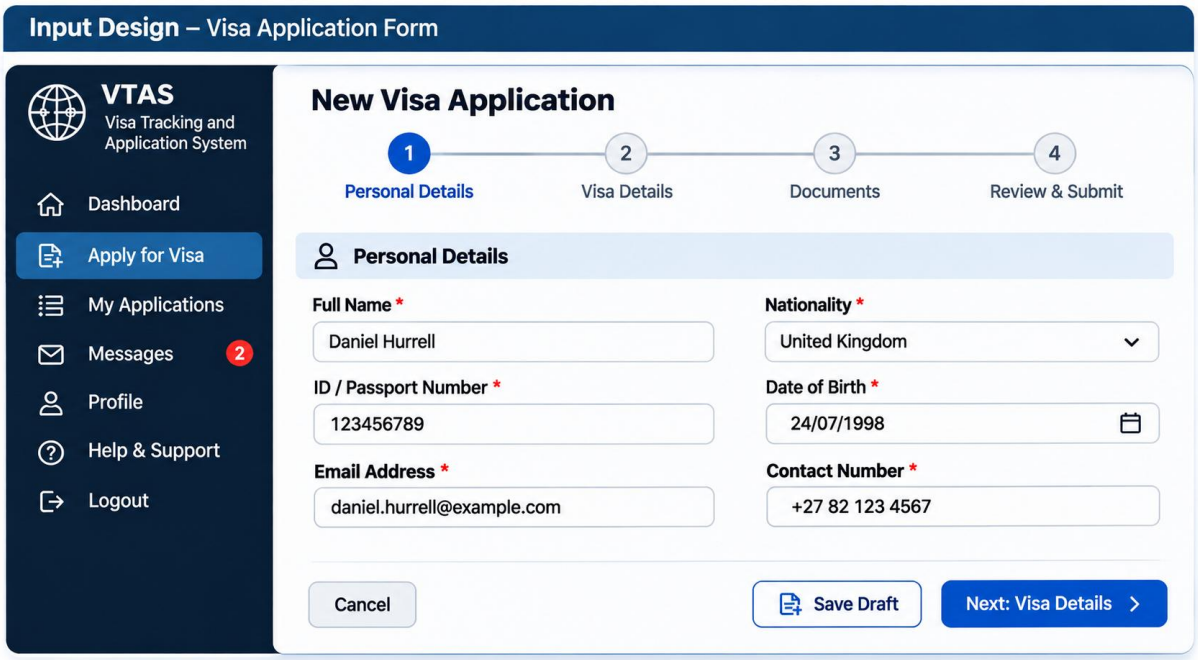
END Generate Report

4.7 Interface Design

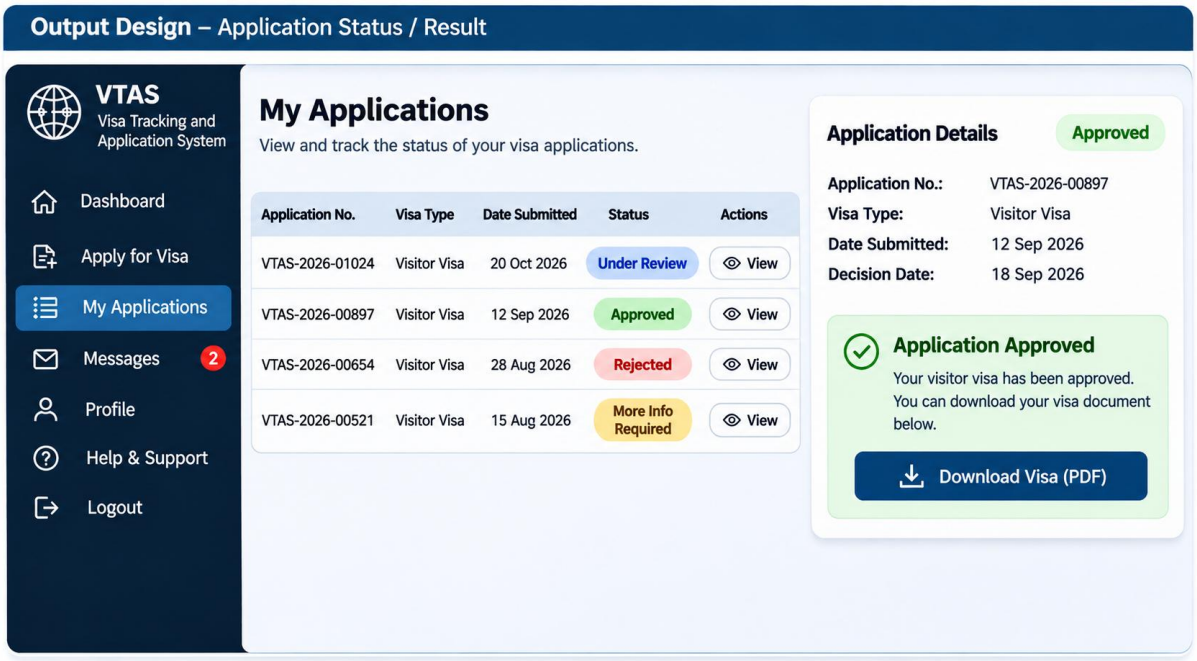
4.7.1 Menu Interface Design



4.7.2 Input Interface Design



4.7.3 Output Interface Design



4.8 Security and Backup Design (Software Concerns)

VTAS will process personal, passport and visa information and must therefore be designed to preserve confidentiality, integrity and availability. Security controls will be built into the software rather than added only after development. The design will also support POPIA principles by limiting access, collecting only required information, recording significant activities and retaining data only for an approved period. POPIA establishes conditions for lawful processing, including minimality, retention and security safeguards (Republic of South Africa, 2013).

4.8.1 Application Security Design

- **Authentication:** Every protected function will require a valid user session. Passwords will be stored as salted cryptographic hashes and never as readable text. (OWASP, 2026).
- **Authorisation:** Role-based access control will restrict applicants to their own records and permit administrators to perform only approved duties. (NIST, n.d.).
- **Session security:** Sessions will use secure cookies, inactivity time-outs and controlled logout. Repeated failed login attempts will be limited and logged.
- **Input validation:** Data will be validated on both the user interface and server. Parameterised database queries will reduce the risk of injection attacks. (OWASP, 2025).
- **Data protection:** HTTPS will protect information in transit, while sensitive fields and backups will be encrypted at rest.
- **Audit logging:** Login events, status changes, record updates, administrative actions and security failures will be logged with the responsible user and date and time.
- **Error handling:** Users will receive clear but non-technical error messages. Detailed errors will be restricted to protected logs and will not expose credentials, database details or internal code.
- **Software maintenance:** Dependencies will be reviewed and updated, secrets will be kept outside source code, and security testing will form part of each development sprint. (OWASP, 2025).

4.8.2 Backup Design

Backup component	Proposed design
Data covered	Database records, uploaded supporting documents, configuration settings, notification templates and protected audit logs.
Backup frequency	Daily incremental backups and a weekly full backup. A backup should also be taken before major releases or database changes.
Storage	Backups will be encrypted and stored separately from the live environment, with at least one protected off-site or logically isolated copy.
Retention	Backup copies will be retained according to the approved operational and legal retention schedule and removed securely when no longer required.
Access	Only authorised technical personnel and controlled service accounts may create, view or restore backups. Access and restore activity will be logged.
Verification	Automated jobs will report success or failure. Restore tests will be performed regularly to confirm that backups are complete, readable and usable. Regular testing and maintenance are core elements of effective contingency planning (NIST, 2010).

4.8.3 Recovery and Continuity

If records are lost or corrupted, the administrator will identify the incident, isolate the affected component and select the most recent verified backup that meets the recovery requirement. The backup will first be restored in a controlled environment, checked for completeness and then returned to service after approval. The proposed recovery point objective is 24 hours, meaning no more than one day of accepted transactions should be lost under the normal daily backup schedule. The proposed recovery time objective is four hours for restoration of the core application and database, subject to the hosting environment and severity of the incident. Failed notification jobs can be reprocessed from the notification records after recovery. The use of defined recovery priorities, restoration procedures and exercises is consistent with NIST contingency-planning guidance (NIST, 2010).

4.8.4 Key Software Security and Backup Concerns

- Weak passwords, stolen sessions or excessive administrator privileges could allow unauthorised access to personal information.
- Incorrect access-control logic could expose one applicant’s records to another user.
- Unvalidated input, outdated packages or insecure interfaces could create application vulnerabilities.
- Hard-coded credentials or poorly protected configuration files could expose database and messaging-service access.
- A backup stored only with the live system could be lost during hardware failure, malware infection or accidental deletion.
- Backups that are not tested may appear successful but fail when a restoration is required.
- Long retention of personal information could increase privacy risk and conflict with approved retention requirements.
- Security patches, backup failures and unusual audit events must therefore be monitored and addressed promptly.

References

National Institute of Standards and Technology (NIST). (2010). *SP 800-34 Rev. 1: Contingency Planning Guide for Federal Information Systems*. NIST Computer Security Resource Center.

National Institute of Standards and Technology (NIST). (n.d.). *Role-based access control*. NIST Computer Security Resource Center Glossary.

OWASP Foundation. (2025). *Application Security Verification Standard, Version 5.0.0*. OWASP ASVS.

OWASP Foundation. (2026). *Password Storage Cheat Sheet*. OWASP Cheat Sheet Series.

Republic of South Africa. (2013). *Protection of Personal Information Act 4 of 2013*. South African Government.

www.mysql.com. (n.d.). *MySQL :: MySQL workbench: Visual database design*. [online] Available at: https://www.mysql.com/products/workbench/design/

dev.mysql.com. (n.d.). *MySQL :: MySQL workbench manual :: 9 Database design and modeling*. [online] Available at: https://dev.mysql.com/doc/workbench/en/wb-data-modeling.html

Ideamerit team (2021). *MySQL database design*. [online] Datensen. Available at: https://www.datensen.com/products/luna-modeler/solutions/mysql/mysql-database-design/

UniversalClass.com. (n.d.). *Database design in MySQL*. [online] Available at: https://www.universalclass.com/articles/computers/database-design-in-mysql.htm

GeeksforGeeks (2022). *MySQL introduction*. [online] GeeksforGeeks. Available at: https://www.geeksforgeeks.org/mysql/what-is-mysql/

Erickson, J. (2024). *MySQL: Understanding what it is and how it's used*. [online] Oracle.com. Available at: https://www.oracle.com/africa/mysql/what-is-mysql/

Node.js (2024). *About*. [online] Node.js. Available at: https://nodejs.org/en/about .